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SARDAR PATEL INSTITUTE OF TECHNOLOGY
Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai – 400058-India
Department of Computer Engineering

Name	Siddh Sakariya
UID no.	2024300200
Experiment No.	01

AIM:	Use Networking utilities
Program 1	
PROBLEM STATEMENT :	PING To test the reachability of a remote host and measure the round-trip time for packets sent over the network using the ICMP protocol.
PROGRAM:	cornell.edu ping cornell.edu uw.edu ping uw.edu ox.ac.uk ping -c 4 ox.ac.uk uchicago.edu ping -c 4 uchicago.edu berkeley.edu ping -c 4 berkeley.edu u-tokyo.ac.jp ping -c 4 www.u-tokyo.ac.jp
RESULT: Function: ● Sends ICMP Echo Request packets to a destination host ● Waits for Echo Reply	



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- Measures round-trip time (RTT)

Observation:

- Whether the host is reachable
- Time delay (latency)
- Packet loss
- TTL value (hop count indication)

cornell.edu

```
siddh.s7@Siddhs-MacBook-Air ~ % ping cornell.edu
PING cornell.edu (128.253.173.246): 56 data bytes
Request timeout for icmp_seq 0
Request timeout for icmp_seq 1
Request timeout for icmp_seq 2
Request timeout for icmp_seq 3
^C
--- cornell.edu ping statistics ---
5 packets transmitted, 0 packets received, 100.0% packet loss
siddh.s7@Siddhs-MacBook-Air ~ %
```

uw.edu

```
siddh.s7@Siddhs-MacBook-Air ~ % ping uw.edu
PING uw.edu (128.208.60.183): 56 data bytes
Request timeout for icmp_seq 0
Request timeout for icmp_seq 1
Request timeout for icmp_seq 2
Request timeout for icmp_seq 3
^C
--- uw.edu ping statistics ---
5 packets transmitted, 0 packets received, 100.0% packet loss
siddh.s7@Siddhs-MacBook-Air ~ %
```

ox.ac.uk

```
siddh.s7@Siddhs-MacBook-Air ~ % ping -c 4 ox.ac.uk
PING ox.ac.uk (35.189.126.202): 56 data bytes
64 bytes from 35.189.126.202: icmp_seq=0 ttl=102 time=304.800 ms
64 bytes from 35.189.126.202: icmp_seq=1 ttl=102 time=322.644 ms
64 bytes from 35.189.126.202: icmp_seq=2 ttl=102 time=251.660 ms
64 bytes from 35.189.126.202: icmp_seq=3 ttl=102 time=259.064 ms

--- ox.ac.uk ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 251.660/284.542/322.644/29.969 ms
siddh.s7@Siddhs-MacBook-Air ~ %
```

uchicago.edu

```
siddh.s7@Siddhs-MacBook-Air ~ % ping -c 4 uchicago.edu
PING uchicago.edu (3.215.148.80): 56 data bytes
Request timeout for icmp_seq 0
Request timeout for icmp_seq 1
Request timeout for icmp_seq 2

--- uchicago.edu ping statistics ---
4 packets transmitted, 0 packets received, 100.0% packet loss
siddh.s7@Siddhs-MacBook-Air ~ %
```

berkeley.edu



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```
siddh.s7@Siddhs-MacBook-Air ~ % ping -c 4 berkeley.edu
PING berkeley.edu (141.193.213.20): 56 data bytes
64 bytes from 141.193.213.20: icmp_seq=0 ttl=55 time=31.427 ms
64 bytes from 141.193.213.20: icmp_seq=1 ttl=55 time=37.140 ms
64 bytes from 141.193.213.20: icmp_seq=2 ttl=55 time=29.169 ms
64 bytes from 141.193.213.20: icmp_seq=3 ttl=55 time=30.788 ms

--- berkeley.edu ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 29.169/32.131/37.140/3.007 ms
siddh.s7@Siddhs-MacBook-Air ~ %
```

www.u-tokyo.ac.jp

```
siddh.s7@Siddhs-MacBook-Air ~ % ping -c 4 www.u-tokyo.ac.jp
PING www.u-tokyo.ac.jp (210.152.243.234): 56 data bytes
Request timeout for icmp_seq 0
Request timeout for icmp_seq 1
Request timeout for icmp_seq 2

--- www.u-tokyo.ac.jp ping statistics ---
4 packets transmitted, 0 packets received, 100.0% packet loss
siddh.s7@Siddhs-MacBook-Air ~ %
```

Program 2

PROBLEM STATEMENT :

NSLOOKUP

To query the Domain Name System (DNS) and obtain the IP address or other DNS records associated with a given domain name.

PROGRAM:

cornell.edu

nslookup cornell.edu

uw.edu

nslookup uw.edu

ox.ac.uk

nslookup www.ox.ac.uk

uchicago.edu

nslookup uchicago.edu

berkeley.edu

nslookup berkeley.edu

u-tokyo.ac.jp



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nslookup www.u-tokyo.ac.jp

RESULT:

Function

- Queries DNS servers
- Resolves domain names to IP addresses
- Can retrieve different DNS records (A, MX, NS, etc.)

Observation

- IP address corresponding to a domain name
- DNS server used for resolution
- Whether DNS resolution is successful

cornell.edu

```
Server:      172.16.10.4
Address:     172.16.10.4#53
```

Non-authoritative answer:

```
Name:   cornell.edu
Address: 128.253.173.244
Name:   cornell.edu
Address: 128.253.173.242
Name:   cornell.edu
Address: 128.253.173.241
Name:   cornell.edu
Address: 128.253.173.243
Name:   cornell.edu
Address: 128.253.173.246
Name:   cornell.edu
Address: 128.253.173.245
```

```
siddh.s7@Siddhs-MacBook-Air ~ %
```

uw.edu

```
siddh.s7@Siddhs-MacBook-Air ~ % nslookup uw.edu
Server:      172.16.10.4
Address:     172.16.10.4#53
```

Non-authoritative answer:

```
Name:   uw.edu
Address: 34.168.51.100
Name:   uw.edu
Address: 34.82.219.4
Name:   uw.edu
Address: 128.208.60.183
Name:   uw.edu
Address: 128.208.60.208
```

```
siddh.s7@Siddhs-MacBook-Air ~ %
```

www.ox.ac.uk



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```
siddh.s7@Siddhs-MacBook-Air ~ % nslookup www.ox.ac.uk
Server:      172.16.10.4
Address:     172.16.10.4#53

Non-authoritative answer:
www.ox.ac.uk canonical name = www.ox.ac.uk.cdn.cloudflare.net.
Name:   www.ox.ac.uk.cdn.cloudflare.net
Address: 172.66.169.161
Name:   www.ox.ac.uk.cdn.cloudflare.net
Address: 104.20.34.13

siddh.s7@Siddhs-MacBook-Air ~ %
```

uchicago.edu

```
siddh.s7@Siddhs-MacBook-Air ~ % nslookup uchicago.edu
Server:      172.16.10.4
Address:     172.16.10.4#53

Non-authoritative answer:
Name:   uchicago.edu
Address: 3.215.148.80

siddh.s7@Siddhs-MacBook-Air ~ %
```

berkeley.edu

```
siddh.s7@Siddhs-MacBook-Air ~ % nslookup berkeley.edu
Server:      172.16.10.4
Address:     172.16.10.4#53

Non-authoritative answer:
Name:   berkeley.edu
Address: 141.193.213.21
Name:   berkeley.edu
Address: 141.193.213.20

siddh.s7@Siddhs-MacBook-Air ~ %
```

www.u-tokyo.ac.jp

```
siddh.s7@Siddhs-MacBook-Air ~ % nslookup www.u-tokyo.ac.jp
Server:      172.16.10.4
Address:     172.16.10.4#53

Non-authoritative answer:
Name:   www.u-tokyo.ac.jp
Address: 210.152.243.234

siddh.s7@Siddhs-MacBook-Air ~ %
```

Program 3

**PROBLEM
STATEMENT:**

IFCONFIG

To display and configure network interface parameters such as IP address, netmask, MAC address, and interface status on a system.

PROGRAM:

ifconfig



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RESULT:

Function

- Displays network interface configuration
- Shows:
- IP address
- Netmask
- MAC address
- Interface status (UP/DOWN)

Observation

- Active and inactive interfaces
- Assigned IP addresses
- Network interface parameters



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```
siddh.s7@Siddhs-MacBook-Air ~ % ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
    options=1203<RXCSUM, TXCSUM, TXSTATUS, SW_TIMESTAMP>
    inet 127.0.0.1 netmask 0xff000000
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
    nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anp11: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether f6:fe:6d:be:df:85
    media: none
    status: inactive
anp10: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether f6:fe:6d:be:df:84
    media: none
    status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether f6:fe:6d:be:df:64
    nd6 options=201<PERFORMNUD,DAD>
    media: none
    status: inactive
en4: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether f6:fe:6d:be:df:65
    nd6 options=201<PERFORMNUD,DAD>
    media: none
    status: inactive
en1: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
    options=460<TSO4,TSO6,CHANNEL_IO>
    ether 36:63:5d:28:e9:40
    media: autoselect <full-duplex>
    status: inactive
en2: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
    options=460<TSO4,TSO6,CHANNEL_IO>
    ether 36:63:5d:28:e9:44
    media: autoselect <full-duplex>
    status: inactive
bridge0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=63<RXCSUM, TXCSUM, TSO4, TSO6>
    ether 36:63:5d:28:e9:40
    Configuration:
        id 0:0:0:0:0:0 priority 0 hellotime 0 fwddelay 0
        maxage 0 holdcnt 0 proto stp maxaddr 100 timeout 1200
        root id 0:0:0:0:0:0 priority 0 ifcost 0 port 0
        ipfilter disabled flags 0x0
    member: en1 flags=3<LEARNING,DISCOVER>
        ifmaxaddr 0 port 8 priority 0 path cost 0
    member: en2 flags=3<LEARNING,DISCOVER>
        ifmaxaddr 0 port 9 priority 0 path cost 0
    nd6 options=201<PERFORMNUD,DAD>
    media: <unknown type>
    status: inactive
utun0: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
    inet6 fe80::e107:e7c5:dd68:aa3c%utun0 prefixlen 64 scopeid 0xd
    nd6 options=201<PERFORMNUD,DAD>
ap1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
    ether 56:d2:4a:67:81:ba
    nd6 options=201<PERFORMNUD,DAD>
    media: autoselect (none)
    status: inactive
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
    ether 92:a4:01:a5:59:52
    inet6 fe80::102b:9008:c052:46ef%en0 prefixlen 64 secured scopeid 0xb
    inet 10.10.72.173 netmask 0xfffff00 broadcast 10.10.72.255
    nd6 options=201<PERFORMNUD,DAD>
    media: autoselect
    status: active

siddh.s7@Siddhs-MacBook-Air ~ % ifconfig -l

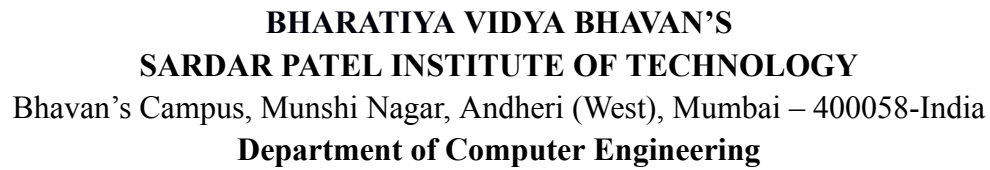
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
    options=1203<RXCSUM, TXCSUM, TXSTATUS, SW_TIMESTAMP>
    inet 127.0.0.1 netmask 0xff000000
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
    nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anp11: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether a2:ef:55:52:f4:11
    inet6 fe80::a0ef:55ff:fe52:f411%anp11 prefixlen 64 scopeid 0xf
    nd6 options=201<PERFORMNUD,DAD>
    media: autoselect (none)
utun1: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::dffa:846d:f01d:c2c5%utun1 prefixlen 64 scopeid 0x10
    nd6 options=201<PERFORMNUD,DAD>
utun2: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 2000
    inet6 fe80::59b4:f9ed:213:4d4d%utun2 prefixlen 64 scopeid 0x11
    nd6 options=201<PERFORMNUD,DAD>
utun3: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1000
    inet6 fe80::ce81:b1c:bd2c:69e%utun3 prefixlen 64 scopeid 0x12
    nd6 options=201<PERFORMNUD,DAD>
siddh.s7@Siddhs-MacBook-Air ~ % ifconfig -l

lo0 gif0 stf0 anp11 anp10 en3 en4 en1 en2 bridge0 utun0 ap1 en0 awdl0 llw0 utun1 utun2 utun3
siddh.s7@Siddhs-MacBook-Air ~ %
```



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Program 4	
PROBLEM STATEMENT:	NETSTAT To display network connections, routing tables, interface statistics, and protocol-wise network information of the system.
PROGRAM:	netstat -t -n -l
RESULT: Function ● Displays network statistics ● Shows: ● Active connections ● Listening ports ● Routing table ● Protocol-wise statistics Observation ● Established network connections ● Open and listening ports ● Network protocol usage	



Program 5

TRACEROUTE

To trace the path taken by packets from the local system to a remote destination and determine the number of intermediate routers (hops) along the route.

```
traceroute math.hws.edu
traceroute www.hws.edu
```

RESULT:



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Function

- Sends packets with increasing TTL values
- Each router that drops a packet replies with ICMP Time Exceeded
- Reveals the route + delay at each hop

Observation

- Sequence of intermediate routers
- IP address / hostname of each hop
- Time taken to reach each hop
- Total hop count to destination

math.hws.edu

```
siddh.s7@Siddhs-MacBook-Air ~ % traceroute math.hws.edu
traceroute to math.hws.edu (64.89.144.237), 64 hops max, 40 byte packets
 1 10.10.72.1 (10.10.72.1) 4.093 ms 3.743 ms 4.467 ms
 2 194.0.0.1 (194.0.0.1) 3.209 ms 3.292 ms 3.333 ms
 3 103.104.226.57 (103.104.226.57) 4.450 ms 3.640 ms 4.381 ms
 4 43.252.192.229 (43.252.192.229) 3.394 ms 3.511 ms *
m 5 * * *
 6 172.28.118.237 (172.28.118.237) 9.203 ms
 27.109.1.149 (27.109.1.149) 4.805 ms 3.748 ms
 7 223-30-0-0.lan.sify.net (223.31.147.249) 4.684 ms 6.255 ms *
 8 * 100.72.69.142 (100.72.69.142) 5.761 ms *
 9 * 100.72.19.10 (100.72.19.10) 5.641 ms *
10 * * *
11 * 47.247.153.57 (47.247.153.57) 11.397 ms *
12 * * 103.198.140.174 (103.198.140.174) 8.861 ms
13 103.198.140.202 (103.198.140.202) 196.294 ms
be3184.ccr42.par01.atlas.cogentco.com (154.54.38.157) 131.348 ms *
14 port-channel2732.ccr92.lhr01.atlas.cogentco.com (154.54.75.138) 180.231 ms 179.290 ms 179.418 ms
15 be23.bngnyhecr0.ip.firstlight.net (66.152.97.50) 199.625 ms 190.020 ms 195.055 ms
16 be3467.ccr32.bos01.atlas.cogentco.com (154.54.94.45) 217.050 ms 213.022 ms
be30.bngnyheer1.ip.firstlight.net (66.152.98.226) 200.768 ms
17 38.104.52.170 (38.104.52.170) 196.694 ms
216-179-84-82.static.firstlight.net (216.179.84.82) 197.506 ms
38.104.52.170 (38.104.52.170) 202.275 ms
18 * * *
19 * * *
20 * be30.bngnyheer1.ip.firstlight.net (66.152.98.226) 295.899 ms *
21 be30.bngnyheer1.ip.firstlight.net (66.152.98.226) 319.935 ms
216-179-84-82.static.firstlight.net (216.179.84.82) 201.511 ms *
```

www.hws.edu



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Department of Computer Engineering

```
siddh.s7@Siddhs-MacBook-Air ~ % traceroute www.hws.edu
traceroute to www.hws.edu (209.43.55.179), 64 hops max, 40 byte packets
 1 10.10.72.1 (10.10.72.1) 4.093 ms 4.149 ms 3.232 ms
 2 194.0.0.1 (194.0.0.1) 3.990 ms 3.027 ms 3.365 ms
 3 103.104.226.57 (103.104.226.57) 4.478 ms 3.631 ms
   static-5.21.248.49-tataidc.co.in (49.248.21.5) 5.426 ms
 4 10.129.10.226 (10.129.10.226) 4.667 ms * *
 5 * * *
 6 * 172.28.118.237 (172.28.118.237) 5.636 ms
   27.109.1.149 (27.109.1.149) 5.104 ms
 7 223-30-0-0.lan.sify.net (223.31.147.249) 7.401 ms * *
 8 * * *
 9 * * *
10 * 100.72.19.13 (100.72.19.13) 68.041 ms 63.621 ms
11 if-bundle-22-2.qcore1.pvu-paris.as6453.net (80.231.154.199) 128.660 ms *
   if-bundle-15-2.qcore1.pye-paris.as6453.net (80.231.154.32) 149.289 ms
12 if-bundle-22-2.qcore1.pvu-paris.as6453.net (80.231.154.199) 145.921 ms * *
13 80.231.245.15 (80.231.245.15) 133.835 ms * *
14 ae-23.a02.sngpsi07.sg.bb.gin.ntt.net (131.103.117.14) 292.089 ms 305.966 ms 239.069 ms
15 * * *
16 ae-15.a03.sngpsi07.sg.bb.gin.ntt.net (129.250.6.65) 311.274 ms
   ae-18.a03.sngpsi07.sg.bb.gin.ntt.net (129.250.6.73) 296.604 ms
   atl-bb1-link.ip.twelve99.net (62.115.138.71) 203.580 ms
17 * ae11.edge3.sng3.sp.lumen.tech (4.68.62.129) 64.596 ms
   atl-b24-link.ip.twelve99.net (62.115.143.237) 245.501 ms
18 dls-b7-link.ip.twelve99.net (62.115.116.212) 296.585 ms * *
19 dls-b24-link.ip.twelve99.net (62.115.139.131) 331.122 ms 243.306 ms 240.831 ms
20 databank-ic-370712.ip.twelve99-cust.net (213.248.99.199) 239.687 ms 239.182 ms 263.262 ms
21 212-69-156-186.databank.com (212.69.156.186) 307.230 ms 242.309 ms 241.976 ms
22 212-69-156-186.databank.com (212.69.156.186) 246.247 ms
   212-69-156-181.databank.com (212.69.156.181) 291.207 ms
   212-69-156-186.databank.com (212.69.156.186) 282.718 ms
23 212-69-156-181.databank.com (212.69.156.181) 248.980 ms
   212-69-156-186.databank.com (212.69.156.186) 249.494 ms
   212-69-157-101.databank.com (212.69.157.101) 261.559 ms
24 212-69-157-101.databank.com (212.69.157.101) 284.340 ms 306.054 ms *
25 212-69-157-101.databank.com (212.69.157.101) 318.003 ms * *
```

CONCLUSION:

From the performed experiments, it is concluded that various network diagnostic commands such as ping, nslookup, ifconfig, netstat, and traceroute are essential tools for analyzing and troubleshooting computer networks.

The ping command verifies network connectivity and measures latency, nslookup ensures proper DNS name resolution, ifconfig displays and confirms network interface configuration, netstat monitors active network connections and protocol statistics, and traceroute identifies the routing path and intermediate hops between source and destination.

Together, these commands provide a comprehensive understanding of network status, performance, configuration, and routing, thereby helping in effective network monitoring and troubleshooting.